

ECM3412

UNIVERSITY OF EXETER

**FACULTY OF ENVIRONMENT, SCIENCE
AND ECONOMY**

COMPUTER SCIENCE

Examination, May 2023

Nature Inspired Computation

Module Leader: Zeliang Wang

Duration: TWO HOURS

No word count specified

Answer ALL questions.

The marks for this module are calculated from 60% of the percentage mark for this paper plus 40% of the percentage mark for associated coursework.

Candidates are permitted to use *non-programmable* portable electronic calculators in this examination.

This is a CLOSED BOOK examination.

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Question 1

- (a) What are the differences between exact and approximate algorithms? Give an example for each type of algorithm.

(6 marks)

- (b) In evolutionary algorithms, encoding is the way to represent a problem in the form of a chromosome. State the advantages and disadvantages between direct and indirect encoding.

(7 marks)

- (c) What are the properties of the swarm intelligence algorithm family? Name two swarm intelligence algorithms.

(6 marks)

- (d) Briefly describe the main algorithmic elements of *particle swarm optimisation (PSO)*. In your answer, you should include feasible parameter settings for each of the parameters associated with the algorithm.

(7 marks)

- (e) Flocking behaviour is commonly seen in animal societies. Give three examples of flocking behaviour, and briefly explain the benefits.

(6 marks)

- (f) What are the properties of Artificial Immune Systems (AIS) as applied to computer security problems? Briefly discuss four properties.

(8 marks)

(Total 40 marks)

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Question 2

- (a) Briefly explain the hill climbing algorithm. Describe two improvements on it.

(8 marks)

- (b) In evolutionary algorithms, a selection procedure is used to choose candidates for mutations and crossover. Explain tournament selection and discuss the selection pressure for those cases highlighting the effect on the algorithm.

- tournament size 5 in population size 5000
- tournament size 5 in population size 20

(9 marks)

- (c) An evolutionary algorithm chromosome has 6 binary-encoded genes, denoted: $b_0b_1b_2b_3b_4b_5$. The fitness function of the chromosome is calculated as the number of leading ones starting from b_0 , e.g., $F(111001) = 3$ and $F(010010) = 0$.

- (i) For the following chromosomes, complete this table to calculate the probabilities for selection using a roulette wheel selection. Show your workings.

chromosome	fitness	probability
110010		
111001		
101010		
111100		

(4 marks)

- (ii) Select 2 parents for the crossover function. Show your workings.

If you need random numbers, choose in order from the following:

.86, .23, .67, .14, .34, .08, .11, .29, .85, .76, .43, .47, .89, .80, .98, .58, .03, .57, .49, .92

Repeat from the beginning if you need more.

(5 marks)

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- (iii) Generate the new offspring based on the selected parents from (ii). The crossover function is a uniform crossover with this binary random mask 101010. Show your workings.

(4 marks)

(Total 30 marks)

Question 3

- (a) Multi-objective evolutionary algorithms (MOEA) can be used for many real world problems where there is a trade-off between objectives. What is meant by the term Pareto-front in MOEA? Give an example of Pareto-front.

(6 marks)

- (b) Self-organising maps can be applied to unsupervised learning tasks. Explain what is meant by the term *unsupervised* in this context and how *unsupervised learning* differs from *supervised learning*.

(5 marks)

- (c) For a binary classification problem, a set of training data is given in the following table

Input (x_1)	Input (x_2)	Target (y)
0	2	0
-1	1	0
-1	-1	1
1	1	1

- (i) Sketch a linear decision boundary that separates these two different classes in y .

(4 marks)

- (ii) Draw a single-neuron perceptron diagram to solve this classification problem and write down an appropriate activation function, with possible weight values that can obtain a mapping from the input (x_1, x_2) to the target output y .

(7 marks)

- (iii) Validate your solution for the above question with all four input patterns, i.e., present each input once to the perceptron model and calculate the output. Write down the steps of your calculations.

(8 marks)**(Total 30 marks)**